

---

## PH 626: Elementary Theoretical Particle Physics

---

|               |                                                                       |
|---------------|-----------------------------------------------------------------------|
| Course Code   | PH 626                                                                |
| Course Name   | Elementary Theoretical Particle Physics                               |
| L-T-P-C       | 3-0-0-3                                                               |
| Prerequisites | PH612 (Nuclear and Particle Physics), PH301/PH513 (Quantum Mechanics) |
| Intended for  | UG/PG/I-PhD/PhD                                                       |
| Approval      | 52nd BoA (Nov 2023)                                                   |

---

### Course Contents

- **Feynman Calculus:** Decays, scattering and cross-sections, Mandelstem variables, Fermi Golden rule, Golden rule for two particle decays and scattering of particles, two-body scattering in the COM frame. Feynman rules and diagrams for a toy theory. (6 Lectures)
- **Quantum Electrodynamics:** Dirac equation, solutions to the Dirac equation, and bilinear covariants, photon, Feynman rules for QED and examples, Casimir's Trick, cross-sections and lifetimes, and renormalization, hadron production in  $e+e-$  collisions, elastic electron-proton scattering. (10 Lectures)
- **Quantum Chromodynamics:** Feynman rules for Chromodynamics, Color factors, quark and antiquark, Pair annihilation in QCD, asymptotic freedom. (8 Lectures)
- **Weak Interactions:** Charged leptonic weak Interactions, decay of muon, neutron, and pion, charged weak interactions of quarks, neutral weak interactions, Electroweak unification and chiral fermion states, Weak isospin and hypercharge, Electroweak mixing. (10 Lectures)
- **Gauge Theories:** Lagrangian formulation of classical particle mechanics and Lagrangians in relativistic field theory, Local gauge invariance and Yang-Mills Theory, Chromodynamics, Feynman rules and Mass term Spontaneous symmetry-breaking, Higgs Mechanism. (8 Lectures)

### Text Books

1. David Griffiths, **Introduction to Elementary Particles**, 2nd edition, Wiley, 2008.
2. F. Halzen and A. D. Martin, **Quarks and Leptons**, John Wiley, 2016.

### References

1. M. Thomson, **Modern Particle Physics**, Cambridge University Press India, 2016.
2. M. E. Peskin, **An Introduction to Quantum Field Theory**, Westview Press, 1995.
3. D. H. Perkins, **Introduction to High Energy Physics**, 4th Edition, Cambridge, 2000.